Test Report No. 2018 – 0484

Agricultural Machinery – Power-Operated Corn Sheller LAKAS KULIGLIG LK-300 STD

College of Engineering and Agro-Industrial Technology, University of the Philippines Los Baños, College, Laguna 4031
Email: amtec.uplb@up.edu.ph Facebook: www.facebook.com/amtec.uplb Website: www.amtec.ceat.uplb.edu.ph
Contact Numbers: (+63) 921 400 7137 (Smart) or (+63) 977 204 7236 (Globe)

Test Report Number	2018 – 0484
Valid Until	July 2, 2023
Agricultural Machinery	Power-Operated Corn Sheller
Type	Axial flow, peg-tooth open cylinder with sifter and fan, husker/sheller
Brand	LAKAS KULIGLIG
Model	LK-300 STD
Test Requested by	P.I. FARM PRODUCTS, INC. 35 Alarcon St., Sitio Malabo Maysan, Valenzuela City

SUMMARY

Performance Criteria	Requirement as per PAES 208:2000	AMTEC Test
Shelling recovery, %, minimum	97.0	98.9
Shelling efficiency, %, minimum	99.5	100
Losses, %, maximum		
a) Blower loss	1.0	0.25
b) Separation loss	1.0	0.69
c) Unshelled loss	0.5	0
d) Scattering loss	0.5	0.13
Purity, %, minimum	98	99.5
Mechanically damaged kernel, %, maximum	3.0	1.93
Net cracked kernel, %, maximum	5.0	4.33
Noise level, dB(A), maximum	100*	98.7

Note: *Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983.

OBSERVATIONS

1. Manufacturer	P.I. FARM PRODUCTS, INC.
2. Operator's manual	Not Provided
3. Previous valid test reports of similar brand and model	None
4. Prime mover used	8.95 kW MITSUSAN MS120 Diesel Engine
5. Transportation of the machine	Provided with hitch-point, towing bars, and pneumatic tires.

OBSERVATIONS (Cont'n)

6. Adjustments such as belt tensions, clearance, air velocity and others	Belt tension can be adjusted.
7. Presence of grains that are blown back at the feeding port during shelling operation	Occasional
8. Cleaning of the cylinder and concave	Concave can be opened for cleaning
9. Cleaning of the fan and housing assembly	Fan can be accessed and cleaned.
10. Number of operators	Six (two for hauling of test materials to the sheller; one for feeding of test materials; one for mixing and recycling of small corn cobs; and two for bagging)
11. Failure or abnormalities that shall be observed on the sheller or its component parts during and after the shelling operation	No breakdown occurred during the test.
12. Other remarks	Machine can be used as Rice Thresher.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	May 25, 2018
Location of Test	Brgy. Bambang, Candaba, Pampanga
Items Tested	Shelling capacity, Shelling recovery, Speed of rotating components Shelling efficiency, Quality of work, Noise level, Fuel consumption, and Number of operators

METHODS OF TEST

The corn sheller was tested based on the PAES 209:2000 Agricultural Machinery – Power-Operated Corn Sheller - Methods of Test.

DESCRIPTION OF THE MACHINE

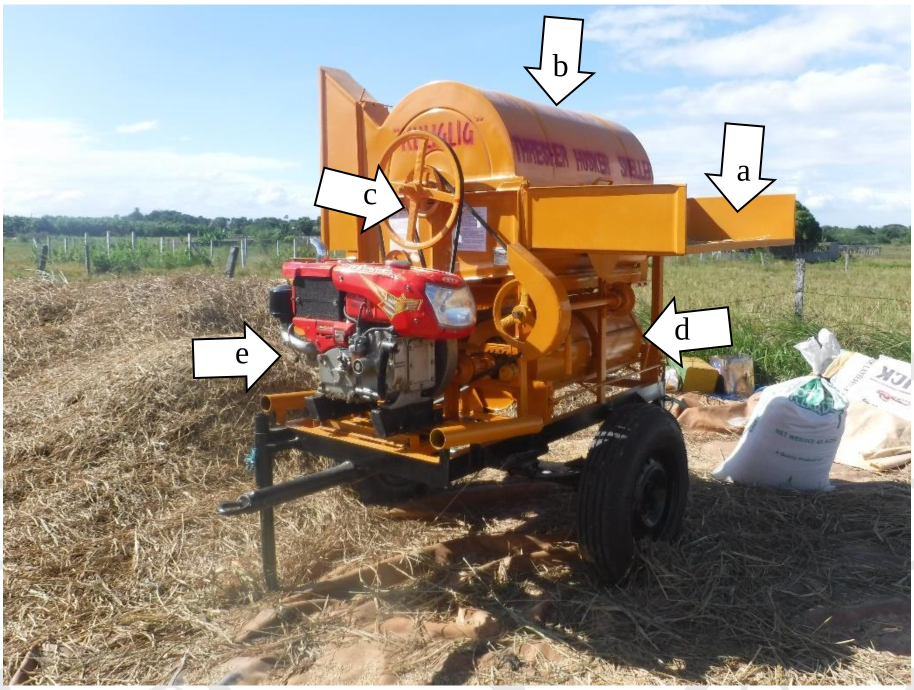

Figure 1. Parts of the Corn Sheller LAKAS KULIGLIG LK-300 STD powered by a 8.95 kW MITSUSAN MS120 Diesel Engine include the following: (a) Feeding table, (b) Shelling cylinder, (c) Belt-pulley transmission, (d) Blower, and (e) Prime mover.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification[b]
1	**Dimensions and weight**		
1.1	Overall length, mm	ND	2230
1.2	Overall width, mm	ND	1560
1.3	Overall height, mm	ND	1900
1.4	Overall weight without engine, kg	ND	NM
2	**Rated capacity, kg/h**	ND	2213
3	**Component speeds (without load)**		
3.1	Cylinder		
3.1.1	Shaft speed, rpm	ND	2580
3.1.2	Cylinder peripheral speed, m/min	ND	964
3.2	Fan or blower shaft, rpm	ND	910
3.3	Oscillating screen shaft, rpm	ND	507
3.4	Auger shaft, rpm	ND	351
4	**Engine**		
4.1	Brand	MITSUSAN	MITSUSAN
4.2	Model	MS120	MS120
4.3	Serial number	ND	ND
4.4	Make	Thailand	Thailand
4.5	Type	ND	Four stroke cycle, water-cooled, horizontal single cylinder, diesel engine
4.6	Rated speed, rpm	2400	2400
4.7	Rated power, kW	8.95	8.95
4.8	Weight, kg	ND	NM
4.8	Starting system	ND	Manual cranking lever
5	**Transmission system**		
5.1	Engine to Shelling cylinder	ND	V-belt and pulley
5.1.1	Engine pulley	ND	127 mm x 2B x 31 mm
5.1.2	Shelling cylinder pulley	ND	406 mm x 2B x 29 mm
5.1.3	Belt size	ND	B-73

Note: ND – No Data, NM – Not Measured

[a] The requesting party did not submit machine specifications.

[b] Data obtained from actual test, nameplates, and on-site measurements.

SPECIFICATIONS (cont'n)

	Item	Manufacturer's Specification[a]	AMTEC Verification[b]
5.2	Shelling cylinder to Oscillating screen	ND	V-belt and pulley
5.2.1	Shelling cylinder pulley	ND	178 mm x 1B x 29 mm
5.2.2	Oscillating screen pulley	ND	229 mm x 1B x 25 mm
5.2.3	Belt size	ND	B-76
5.3	Fan to Auger	ND	V-belt and pulley
5.3.1	Fan pulley	ND	76 mm x 1B x 25 mm
5.3.2	Auger pulley	ND	254 mm x 1B x 25 mm
5.3.3	Belt size	ND	B-53
6	**Type of clutch system**	NA	NA
7	**Shelling chamber**		
7.1	Cylinder		
7.1.1	Type	ND	Open cylinder
7.1.2	Size, L x D, mm	ND	1030 x 475
7.2	Cylinder teeth		
7.2.1	Type	ND	Peg-tooth
7.2.2	Size, L x W x T, mm	ND	60 x 25 x 12
7.2.3	Number per anchor bar	ND	11 and 10
7.2.4	Distance between teeth, mm	NM	NM
7.2.5	Arrangement	ND	Offset with respect to knife bars on the next anchor bar
7.2.6	Material used	ND	Mild steel
7.2.7	Means of attachment	ND	Welded
7.3	Cylinder cover		
7.3.1	Shape	ND	Half-cylindrical horizontal
7.3.2	Material	ND	B.I. sheet (#18)

Note: ND – No Data, NM – Not Measured

[a] The requesting party did not submit machine specifications.

[b] Data obtained from actual test, nameplates, and on-site measurements.

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification[b]
7.3.3	Louver		
7.3.3.1	Number	ND	6
7.3.3.2	Inclination with respect the vertical axis, degrees	ND	77
7.4	Concave		
7.4.1	Lower concave		
7.4.1.1	Material	ND	Cold-rolled steel (8mm)
7.4.1.2	Spacing between grills	ND	15
7.4.1.3	Clearance between concave and cylinder teeth, mm	ND	22
7.4.2	Upper concave		
7.4.2.1	Material used	ND	Cold-rolled steel (8mm)
7.4.2.2	Spacing between grills, mm	ND	18
8	**Feeding Table**		
8.1	Length, mm	ND	1100
8.2	Width, mm	ND	580
8.3	Height from the ground, mm	ND	1220
8.4	Dimension of feeding port, L x W, mm	ND	NM
8.5	Mode of attachment	ND	Bolted
8.6	Material	ND	B.I. sheet (#18)
9	**Oscillating Screen/Sieve**		
9.1	Dimensions, L x W, mm	ND	880 x 860
9.3	Size of perforations, mm	ND	12
9.4	Length of stroke, mm	ND	12
9.5	Angle of inclination, ^{o}	ND	NM
9.6	Material	ND	Perforated B.I. sheet

Note: ND – No Data, NM – Not Measured

[a] The requesting party did not submit machine specifications.

[b] Data obtained from actual test, nameplates, and on-site measurements.

SPECIFICATIONS (cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification[b]
10	**Fan**		
10.1	Number of units	ND	2
10.2	Fan 1		
10.2.1	Type	ND	Centrifugal, radial
10.2.2	Total length	ND	395
10.2.3	Diameter	ND	330
10.2.4	Number of blades	ND	4
10.2.5	Size of inlet port, mm	ND	NM
10.2.6	Material	ND	B.I. sheet (#20)
10.3	Fan 2		
10.3.1	Type	ND	Centrifugal, radial
10.3.2	Total length	ND	295
10.3.3	Diameter	ND	330
10.3.4	Number of blades	ND	4
10.3.5	Size of inlet port, mm	ND	NM
10.3.6	Material	ND	B.I. sheet (#20)
11	**Outlet chute**		
11.1	Angle of inclination, ^{o}	ND	NM
11.2	Material	ND	ND
11.3	Discharge device	ND	Inclined chute with auger mechanism
12	**Transport device**		
12.1	Type	ND	Pneumatic tires with hitch point and two towing bars
12.2	Size	ND	NM
13	**Chassis material**	ND	NM
14	**Safety device/s, if any**	ND	None
15	**Adjustment/s**	ND	Air intake shutter of fan
16	**Tools available with machine**	ND	None
17	**Other special features**	NA	Machine can be used as Rice thresher.

Note: ND – No Data, NM – Not Measured

[a] The requesting party did not submit machine specifications.

[b] Data obtained from actual test, nameplates, and on-site measurements.

RESULTS

Item		Data	
1	**Crop Condition**		
1.1	Variety	Syngenta 8840	
1.2	Days after harvest	Same day	
1.3	Kernel-corn ear ratio, mm	0.80	
1.4	Grain moisture content, %	17.7	
1.5	Size of corn ear, L x D, mm	292 x 57	
2	**Performance Test**		
2.1	Speed of components	Without Load	With Load
2.1.1	Engine, rpm	2580	2544
2.1.2	Shelling cylinder shaft, rpm	647	633
2.1.3	Fan shaft, rpm	910	894
2.1.4	Oscillating screen shaft, rpm	507	498
2.1.5	Grain auger shaft, rpm	351	345
2.1.6	Shelling cylinder peripheral speed, m/min	964	943
2.2	Fan air velocity, m/sec	3.44	NM
2.3	Noise level, dB(A)	Without Load	With Load
2.3.1	Feeding operator	82.6	98.7
2.3.2	Bagger	86.1	99.6
2.4	Shelling time, min	10.25	
2.5	Shelled kernel, kg	378	
2.6	Actual shelling capacity, kg/h	2213	
2.7	Corrected shelling capacity, kg/h	2266*	
2.8	Fuel time, min	16.17	
2.9	Fuel consumed, mL	425	
2.10	Fuel consumption rate, L/h	1.58	

Note: NM – Not Measured
* Adjusted to 20% grain moisture content and 100% purity of shelled grain.

RESULTS (Cont'n)

Item		Data
3	**Laboratory Analysis**	
3.1	Shelling recovery, %	98.9
3.2	Shelling efficiency, %	100
3.3	Blower loss, %	0.25
3.4	Unshelled loss, %	0
3.5	Separation loss, %	0.69
3.6	Scattering loss, %	0.13
3.7	Purity of shelled kernel, %	99.5
3.8	Total shelling loss, %	1.07
3.9	Before shelling, %	
3.9.1	Cracked	0
3.9.2	Broken	0
3.10	After shelling, %	
3.10.1	Cracked	4.33
3.10.2	Broken	1.93
3.11	Net percent, %	
3.11.1	Cracked	4.33
3.11.2	Broken	1.93

Prepared by:

DARWIN C. ARANGUREN
Test Engineer

Verified by:

ROMULO E. EUSEBIO
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

AUG 22 2019
Date

REMINDER

This test report should not be used for advertisement or for promotion of the product but only for evaluation and verification purposes and should not be reproduced in whole or in part without the permission of AMTEC.

For inquiries regarding the test report, you can visit us at our office, contact us at (+63) 921 400 7137 or (+63) 977 204 7236, send us a message at amtec.uplb@up.edu.ph or visit our official website at www.amtec.ceat.uplb.edu.ph.